

HUM 4747: Legal Issues and Cyber Law

Class 4: New Technologies, AI & Modern IP Challenges

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Outline

- ① The Digital Revolution & IP
- ② AI and Intellectual Property
- ③ Copyright in the Digital Age
- ④ Emerging IP Issues
- ⑤ Software, Code & IP
- ⑥ IP and Bangladesh's Digital Future
- ⑦ Recent Global Updates in IP Law
- ⑧ Course Summary & Exam Preparation

Recap: IP in the Traditional World

Over the last 3 classes, we covered:

- What IP is and why it matters
- Types: Copyright, Trademark, Patent, Design, GI, Trade Secret
- Bangladesh's IP laws, enforcement, and challenges

Today's Big Question

All those laws were designed for a **pre-digital, pre-AI world**.

How do they handle **AI-generated content, deepfakes, NFTs, streaming, memes, and open-source software**?

How Technology Changed Everything

Traditional IP World

- Physical books, vinyl records
- Copying was expensive and slow
- Distribution was limited
- Counterfeiting required factories
- Clear human authorship

Digital IP World

- E-books, MP3s, streaming
- Copying is **free and instant**
- Distribution is **global in seconds**
- Anyone with a computer can copy
- AI can create content — **who is the author?**

The Core Challenge

Technology has made it **trivially easy to copy and distribute** creative works — but the laws haven't caught up.

The AI Revolution — ChatGPT, DALL-E, Midjourney

What's Happening?

- **ChatGPT** (OpenAI) generates text: essays, code, poems, legal documents
- **DALL-E / Midjourney / Stable Diffusion** generate images from text prompts
- **Suno / Udio** generate music
- **GitHub Copilot** auto-generates code
- These AI systems are **trained on billions of copyrighted works**

The Million-Dollar Questions

- ① **Who owns AI-generated content?** The user? The AI company? Nobody?
- ② Is training AI on copyrighted data **fair use or infringement?**
- ③ Can AI be listed as an **inventor** on a patent?

Question 1: Who Owns AI-Generated Content?

Current Legal Position (Most Countries)

- Copyright requires a **human author** — AI cannot be an “author”
- U.S. Copyright Office (2023): AI-generated works **cannot be copyrighted**
- Only **human creative contributions** (e.g., selecting, arranging) may be protectable

Case: Théâtre D'opéra Spatial (2022)

Jason Allen won a **Colorado State Fair art competition** using an image generated by Midjourney AI. The U.S. Copyright Office **denied copyright registration** because the image was “not the product of human authorship.”

Implication for Students

If you generate an essay with ChatGPT, **you don't own the copyright**. And submitting it as your own work raises **academic integrity** and **IP issues**.

Question 2: Is Training AI on Copyrighted Data Legal?

The Controversy

AI models like ChatGPT and Stable Diffusion are trained on **vast datasets of copyrighted text, images, and code** scraped from the internet — often **without permission or payment**.

Major Ongoing Lawsuits (2023–2025):

- **The New York Times vs. OpenAI & Microsoft** — claims ChatGPT reproduces NYT articles verbatim
- **Getty Images vs. Stability AI** — Stable Diffusion trained on millions of copyrighted photos
- **Authors Guild vs. OpenAI** — writers claim their books were used without consent
- **GitHub Copilot Lawsuit** — developers claim their open-source code was used to train a commercial product

Key Legal Debate

Is AI training “**fair use**” (learning from works, like a human would)? Or is it **mass copyright infringement**? Courts haven't decided yet.

Question 3: Can AI Be an Inventor?

The DABUS Case (2020–2023)

- Dr. Stephen Thaler created an AI system called **DABUS** (Device for the Autonomous Bootstrapping of Unified Sentience)
- DABUS autonomously invented a **food container design** and a **light beacon**
- Thaler filed patent applications listing **DABUS as the inventor**

Results across countries:

- **USA, UK, EU:** **Rejected** — only a **natural person** can be an inventor
- **Australia:** Initially accepted, then **overturned on appeal**
- **South Africa:** **Granted** — but South Africa doesn't substantively examine patents

The Gap

Current patent law **assumes human inventors**. As AI becomes more autonomous, this framework may need fundamental rethinking.

AI and IP: Summary of Current State

Issue	Traditional IP Answer	AI Challenge
Who is the author/inventor?	The human creator	AI has no legal personhood
Can output be copyrighted?	Yes, if original	Only if human contribution exists
Is training on data legal?	Need license/permission	“Fair use” debate ongoing
Who is liable for infringement?	The human infringer	User? Developer? No-body?

AI and IP: Summary of Current State(contd.)

Bangladesh Context

Bangladesh's IP laws **do not yet address AI**. However, three major updates came in 2023:

- **Copyright Act, 2023** — stronger digital protections
- **Bangladesh Patents Act, 2023** — 20-year term
- **Bangladesh Industrial Designs Act, 2023** — max 25 years

The Copyright Act 2000 and Patents Act 1911 have been replaced. AI regulation remains a global gap.

Music, Movies & Streaming

The Transformation

- **Napster (1999–2001)**: Free music sharing → sued and shut down
- **iTunes (2003)**: Legal digital music purchases
- **Spotify, Netflix, YouTube (2010s)**: Streaming model — pay for access, not ownership
- Result: Piracy decreased in countries where **legal alternatives were affordable**

Bangladesh Reality

- Many use **torrent sites, Telegram channels** for movies/music
- **Streaming subscriptions** seen as expensive
- Local content creators lose revenue

Did You Know?

YouTube's **Content ID** system scans every upload against **100+ million reference files**. It handles **more copyright claims** than any court system in the world.

IP in Gaming — Multiple Layers

A single video game can involve **multiple types of IP**:

- **Copyright**: Code, artwork, music, story, dialogue
- **Trademark**: Game title, studio logo, character names
- **Patent**: Game mechanics (e.g., Namco's patent on mini-games during loading screens)
- **Trade Secret**: Anti-cheat algorithms, server infrastructure

Famous Cases

- **PUBG vs. Fortnite** (2018): PUBG Corp sued Epic Games for copying its battle royale concept. Case eventually dropped — **game mechanics are hard to patent**.
- **Nintendo vs. MariCar** (2018): A go-kart company in Tokyo let tourists dress as Mario characters. Nintendo won — **character costumes are protected IP**.
- **Game mods**: Many mods use copyrighted assets — a grey area.

Memes, Social Media & Copyright

Are Memes Legal?

- Most memes use **copyrighted images or video clips**
- Technically, this is **copyright infringement**
- In practice, many fall under “**fair use**” / “**fair dealing**” (commentary, parody, satire)
- But **commercial use** of memes can trigger lawsuits

Famous Meme IP Cases

- **Pepe the Frog**: Creator Matt Furie copyrighted the character and successfully sued organizations misusing it
- “**Grumpy Cat**”: The owners trademarked the name and licensed it for merchandise — earning millions
- “**Nyan Cat**”: Sold as an NFT for \$600,000 by the original creator

Rule of thumb: If you're making money from someone else's creation, you likely need permission.

Open Source Software & IP

What is Open Source?

Software whose **source code is freely available** for anyone to view, modify, and distribute — but it still has **copyright and licensing rules**.

Common Open Source Licenses:

License	Key Rule	Examples
MIT License	Very permissive; do anything with credit	jQuery, React, VS Code
GPL (GNU)	Must share modifications as open source	Linux, WordPress
Apache 2.0	Permissive + patent grant	Android, Kubernetes
Creative Commons	For creative works, not just code	Wikipedia, Flickr photos

Important for SWE Students

“Open source” does NOT mean “no copyright.” Violating an open-source license (e.g., not giving attribution) is **still copyright infringement**.

Deepfakes and IP

What are Deepfakes?

AI-generated or manipulated **video, audio, or images** that make someone appear to say or do things they never did. Uses **deep learning** technology.

IP and Legal Issues:

- **Right of publicity:** Using someone's likeness without consent
- **Copyright:** The original video/image is copyrighted
- **Trademark:** Deepfakes can damage brand reputation
- **Defamation:** False portrayal causing reputational harm
- **Fraud:** Deepfake voices used in financial scams

Real Cases

- 2024: A finance worker in Hong Kong was tricked into transferring **\$25 million** after a deepfake video call impersonated the company's CFO
- Multiple celebrities have had **non-consensual deepfake content** created using their likeness

NFTs and Digital Ownership

What is an NFT?

A **Non-Fungible Token** is a unique digital certificate of ownership recorded on a **blockchain**. It can represent art, music, videos, or any digital asset.

IP Issues with NFTs:

- Buying an NFT usually does **NOT transfer copyright** — you own the token, not the IP
- People have **minted NFTs of artworks they don't own** — IP theft
- The **Bored Ape Yacht Club** model: NFT purchase includes **commercial usage rights**
- **No legal framework** in most countries (including Bangladesh) specifically for NFTs

Notable NFT IP Disputes

- **Hermès vs. MetaBirkins** (2023): An artist sold NFTs of “MetaBirkin” bags. Hermès sued for **trademark infringement** and won \$133,000 in damages.

Domain Names & Cybersquatting

What is Cybersquatting?

Registering a **domain name** that is identical or similar to an existing **trademark**, with the intent to **profit from selling it** back to the trademark owner.

Examples:

- Someone registers `samsung-bangladesh.com` before Samsung does
- `walmartsucks.com` — criticism sites raise free speech questions
- Typosquatting: `gooogle.com` (extra 'o') to redirect traffic

Resolution Mechanism

- **UDRP** (Uniform Domain-Name Dispute-Resolution Policy) by WIPO/ICANN
- Faster and cheaper than going to court
- Bangladesh: Disputes handled through regular courts (no specific cybersquatting law)

3D Printing and IP

The Challenge

3D printing allows anyone to **manufacture physical objects at home** from digital designs. This threatens:

- **Patents:** Anyone can replicate a patented product
- **Industrial Designs:** Copy the shape/appearance of products
- **Trademarks:** Print fake branded items
- **Copyright:** Reproduce sculptures, figurines, artistic works

Example

A student downloads a 3D file of a patented mechanical part from Thingiverse, prints it, and sells copies. This violates **multiple IP rights** but is **nearly impossible to detect or enforce**.

Future Concern

As 3D printers become cheaper and more capable, **IP enforcement in the physical world** will face the same challenges that digital piracy created for music and movies.

How Software is Protected

Software Copyright

- Software is protected as “**literary work**” under copyright law
- Bangladesh: Section 2(46) of **Copyright Act, 2000**
- Protects the **source code and object code** (the expression)
- Does NOT protect the **underlying algorithm or idea**

Software Patents

- In some countries (USA), software **methods/algorithms** can be patented
- Controversial: Many argue software patents stifle innovation
- Bangladesh: Patent Act doesn't specifically address software patents

Famous Case: Oracle vs. Google (2021)

Google used Java APIs in Android. Oracle sued for copyright. After 10 years, the **U.S. Supreme Court** ruled Google's use was **fair use** — a landmark decision for the software industry.

Code You Write — Who Owns It?

Employment Context

- Code written as an employee usually belongs to the **employer** (“work for hire”)
- Check your **employment contract** — it often has IP assignment clauses
- Some contracts claim ownership of **all code you write, even personal projects**

Freelance / Independent Context

- Default: Copyright belongs to the **creator** (you)
- Unless the contract **assigns copyright** to the client
- Always have a clear **written agreement**

Student Projects

Code you write for university projects is generally **your IP**, unless the university has specific policies. But using copyrighted code from others (without proper licensing) is infringement.

GitHub Stack Overflow & Attribution

GitHub

- Code on GitHub is **NOT automatically public domain**
- Without a license file, the code is “**all rights reserved**”
- Always check the **LICENSE file**
- **GPL code** cannot be used in proprietary software

Stack Overflow

- All content is licensed under **CC BY-SA 4.0**
- You **must give attribution** when using code from SO
- Modifications must be shared under **same license**
- Simply copy-pasting without credit violates the license

Best Practice for Developers

- ① Always add a **LICENSE file** to your repositories
- ② Read licenses before using others' code
- ③ Give **attribution** even when not strictly required
- ④ When in doubt, **ask for permission**

Bangladesh's ICT Sector and IP

Growth of ICT in Bangladesh

- Separate **Ministry of ICT** created in December 2011
- Bangladesh High-Tech Park Authority established
- ICT Policy 2009 and Copyright Act amended for **software protection**
- Growing software export industry (\$1.3 billion+ in IT/ITES exports)

The Paradox

Bangladesh wants to be a **software exporting country**, but its domestic software piracy rate is **80%+**. You can't sell software globally if you don't protect software at home.

What's Needed

- Effective enforcement of software copyright
- IP awareness for tech startups and developers
- Affordable legal alternatives (subscription models, open source)

Recommendations for Bangladesh's Digital IP Future

① Update IP laws for the digital age:

- Address AI-generated content, digital rights management (DRM)
- Create provisions for online copyright enforcement

② Establish a digital IP infrastructure:

- Online patent/trademark filing
- National IP database accessible to all

③ Include IP in tech education:

- University curricula should cover IP for developers
- NSTP 2011 already recommends this

④ Support the creative economy:

- Collective management organizations for music/film
- Fair royalty systems for digital content

⑤ Join international digital IP treaties:

- WIPO Copyright Treaty (WCT)
- WIPO Performances and Phonograms Treaty (WPPT)

Recent Global Developments (2023–2025)

EU AI Act (2024)

- World's first comprehensive AI regulation
- Requires AI systems to **disclose** when content is AI-generated
- AI companies must provide **summaries of training data** (including copyrighted material)
- Sets obligations based on **risk levels** of AI systems

U.S. Executive Order on AI (October 2023)

- Calls for study of AI's impact on IP
- U.S. Copyright Office: AI output without human creativity **cannot be copyrighted**
- USPTO: AI cannot be named as inventor on patent applications

China's Approach

China allows **copyright for AI-generated content if there is sufficient human involvement** in the creation process (Beijing Internet Court, 2023).

Recent Developments in Bangladesh IP

Positive Steps

- **GI Registration:** Jamdani, Hilsa getting GI recognition
- **DPDT modernization:** Partial automation with WIPO assistance
- **IPR Project:** EC and WIPO-funded project for IP modernization
- **Digital Bangladesh** vision driving tech awareness
- **BIPO proposal:** Unified IP office under discussion

Still Pending

- Modern Patent Act (to replace 1911 Act)
- Trade Secret legislation
- IC Layout Design protection
- Plant Variety Protection law
- Joining Patent Cooperation Treaty (PCT)

Full Course Summary — IP Law in 4 Classes

Class	Topic	Key Points
1	Introduction to IP	Definition, history, WIPO/TRIPS, Bangladesh IP admin, why IP matters
2	Types of IP	Copyright, Trademark, Patent, Design, GI, Trade Secret — features, duration, examples
3	Laws & Enforcement	BD legal framework, penalties, enforcement agencies, challenges, case studies
4	Technology & AI	AI & IP, digital copyright, open source, deepfakes, NFTs, software IP, future

Key Concepts to Remember

- ① IP = Creations of the mind; protected by law to balance **creator rights + public interest**
- ② **6 main types**: Copyright, Trademark, Patent, Industrial Design, GI, Trade Secret
- ③ Bangladesh key laws: **Patents Act 1911, TM Act 2009, Copyright Act 2000, GI Act 2013**
- ④ Two IP offices: **DPDT** (Ministry of Industries) and **Copyright Office** (Ministry of Cultural Affairs)
- ⑤ Three enforcement remedies: **Civil, Criminal, Administrative**
- ⑥ TRIPS gives LDCs like Bangladesh **extended transition periods**
- ⑦ AI-generated content: **generally not copyrightable** (requires human authorship)
- ⑧ “Open source” $\neq$ “No copyright” — **always check the license**
- ⑨ Bangladesh needs: modern laws, digital infrastructure, IP courts, awareness

In 20 years, will IP law as we know it still exist?

Consider:

- AI can generate unlimited content — does **scarcity-based IP** make sense?
- 3D printing can replicate any physical design — can patents survive?
- Open source has built the internet — is **sharing better than protecting**?
- Developing countries struggle to enforce IP — is the system **designed for the rich**?
- As a future SWE professional, how will **you** navigate these issues?

Your Action Items as Future Professionals

- ① Always **license your code** (choose MIT, GPL, Apache, etc.)
- ② **Respect others' IP** — give credit, follow licenses
- ③ **Stay informed** — IP law is evolving rapidly
- ④ **Protect your innovations** — register trademarks for your startup

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Thank You!

Questions?

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Intellectual Property Law Module

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